

SAIGANESH NELLORE

+1 (773) 822-5301 | nsaiganesh2003@gmail.com | Chicago, IL, USA | linkedin.com/in/saiganeshn/ | github.com/SAIGANESH02 | saiganeshn-ai-portfolio.vercel.app/

PROFESSIONAL SUMMARY

Machine Learning Engineer specializing in production LLM systems, high-performance inference, and GPU-accelerated ML infrastructure. Built LLM inference stacks supporting 100K+ daily requests using vLLM, TensorRT, Triton, PyTorch/JAX, quantization, and distributed GPU training, reducing model serving costs by up to 85%. Experienced across agentic AI, model optimization, distributed training, evaluation, and production ML systems.

SKILLS

- ML and GenAI:** LLM, RAG, LLM Evaluation, NLP, Computer Vision, Prompt Engineering, Model Compression, Model Distillation, RLHF/GRPO
- Frameworks:** PyTorch, Transformers, TensorFlow, LangChain, LangGraph, Scikit-learn, NumPy, Pandas
- Serving and MLOps:** vLLM, AWS SageMaker, AWS Bedrock, CI/CD, Docker, Kubernetes, Jenkins, AWS, Apache Spark, Microsoft Azure, GitHub Actions, MLflow, Apache Airflow, Databricks, LLMOps, GCP
- Data and Databases:** SQL, Postgres, MongoDB, Spark, Snowflake, BigQuery, Power BI
- Programming Languages:** Python, C/C++, Java, JavaScript, R

PROFESSIONAL EXPERIENCE

Cincinnatus LLC (Client: Google)
Sr. Machine Learning Engineer (Contract)

May 2026 - Present
Remote

- Serve as MLOps engineer improving frontier LLM performance on ML systems topics - distributed training (FSDP, tensor/pipeline parallelism), training infrastructure, and GPU kernel optimization (Pallas/Triton, JAX/PyTorch).
- Design expert-level ML systems tasks and reference solutions, and author evaluation rubrics assessing training-pipeline design, distributed-systems reasoning, and kernel-level optimization for model training data.
- Evaluate model outputs on training-infrastructure and kernel-optimization problems; collaborate with SMEs to keep evaluation criteria consistent and technically accurate.

XSELL Technologies
Machine Learning Engineer III

May 2025 - April 2026
Chicago, IL, USA

- Built and scaled a production agentic voice platform (STT, TTS, LLM dialog) enabling sub-second patient conversations across thousands of daily users in regulated healthcare environments.
- Fine-tuned and distilled domain LLMs (Llama 3.1, Qwen 3) to deliver faster, cheaper and scaled LLM inference to 100K+ requests/day using vLLM + TensorRT and compression (quantization/distillation), cutting out-of-box spend from \$300/day to \$40/day while preserving quality and safety.
- Scaled LLM training across a multi-node GPU cluster using Accelerate with FSDP and checkpointing, improving throughput and resuming runs cleanly after failures.
- Productionized multi-model serving on AWS (SageMaker, Bedrock, Triton) with autoscaling, observability, GPU utilization tuning, and cost controls to sustain latency and throughput targets.
- Built evaluation + CI/CD (GitHub Actions) with automated labeling, drift checks, and guardrails, plus HIPAA-aligned PHI redaction and secure logging for operations.

Paramount
AI Engineering Intern - Conversational AI

June 2024 - January 2025
Des Plaines, IL, USA

- Led a 3-engineer team to deliver a fully automated Voice AI sales agent for Car dealership sales and service appointment automation which is capable of handling concurrent calls and dynamic conversations without human intervention.
- Tuned OpenAI Realtime voice agent via A/B testing and feedback loops, cutting human-fallback rate by 46% and keeping end-to-end response latency under one second.
- Managed tasks, planning, and cross-team collaboration, integrating Azure services, Whisper, Twilio, Replit, and LangChain with CRM and ML infrastructure for robust components and increased system uptime.

Zoho
AI Engineer

December 2021 - June 2023
Remote

- Improved financial fraud-detection model recall by 30% by encoding domain rules as differentiable constraints in training.
- Built an ML-driven automated database tuning system (OtterTune) that reused historical session data to optimize configuration knobs, achieving up to 80% performance gains over default setups.
- Configured sub-minute DB configuration within 94% of expert setups and improved latency/performance by up to 80% over default settings.

OTHER EXPERIENCE / PROJECTS

ML Engineer @ Vanguard - Harnessing LLMs for Automatic Test Question Generation for RAGs - [Link to project](#)

- Optimized RAG evaluations for a customer representative use case by leveraging LLMs to reduce testing time and lower developer and SME costs.
- Projected \$300K+ annual SME cost savings (validated with Vanguard stakeholders) by automating test-question creation with quality metrics for accuracy, diversity, and relevance.
- Launched an LLM-driven pipeline for RAG testing, generating chunk-based questions that enhance coverage, variety, and real-time insights.
- Applied advanced chunking (line, para, LLM) and specialized prompts to ensure test-question accuracy, diversity, and relevance across documents.

Data Scientist @ Why of AI - RaceGPT: NextLap's Race Radio Intelligence Copilot - [Link to project](#)

- Built an end-to-end race comms intelligence pipeline (denoise, diarization, alignment, transcription, embeddings, RAG) to convert multi-channel radio into citation-grounded NL search and strategy answers.
- Developed competitor-analysis tooling over driver/crew transcripts to surface key calls (cautions, tire, fuel, issues) and reduce manual review for strategists and analysts.
- Architected PoC blueprints and data/LLM workflows to standardize delivery and decision support across race programs, accelerating repeatable AI deployments.

Workflow-Orchestrated RAG Chatbot - [Link to project](#)

- Built an end-to-end RAG agent (n8n, PostgreSQL + PGVector) to automate article ingestion and deliver document-grounded conversational Q&A.
- Designed modular ingestion and chat workflows with clear schemas (metadata, embeddings, chat histories) to improve maintainability, traceability, and extensibility.
- Containerized n8n, Postgres, and Adminer with Docker Compose to simplify deployment and debugging, enabling simplest local setup for rapid iteration.

EDUCATION

Northwestern University
Master's, Artificial Intelligence

September 2023 - December 2024
GPA: 3.95

Amrita School of Engineering
Bachelor's, Computer Science

May 2019 - June 2022
GPA: 3.65